



DIGITAL LENDING: PORTFOLIO OPTIMIZATION

INDUSTRY CONTEXT

Digital lending institutions in emerging markets have witnessed rapid growth by leveraging technology-driven customer acquisition, alternative data sources, and faster credit decisioning. These lenders cater to a diverse customer base across urban and semi-urban regions, including first-time borrowers and underbanked segments, through products such as unsecured personal loans, SME working capital loans, and buy-now-pay-later (BNPL) offerings.

While scale and speed have enabled strong topline growth, evolving borrower behavior, macroeconomic volatility, and rising acquisition costs have increased pressure on portfolio quality and profitability. Many lenders now face the challenge of balancing growth ambitions with prudent risk management, while effectively leveraging the rich data generated across the customer lifecycle.

BUSINESS CHALLENGE

Despite having access to extensive customer, loan, repayment, and behavioral data, lenders often struggle to translate this information into actionable insights for strategic decision-making. Key issues include limited visibility into risk drivers, delayed identification of potential delinquencies, suboptimal pricing and product structures, and fragmented performance monitoring for leadership.

The central challenge is to design a data-driven approach that enables lenders to understand customer risk and value holistically, detect early signs of financial stress, optimize growth strategies without compromising portfolio quality, and support leadership with clear, forward-looking insights.

PROBLEM STATEMENT

How can a digital lending institution leverage end-to-end customer and transaction data to strengthen credit risk assessment, enable early detection of delinquencies, optimize pricing and product strategies, and drive sustainable, risk-adjusted portfolio growth?

KEY QUESTIONS TO ADDRESS

Participants are expected to explore and answer the following:

1. Which customer segments exhibit materially different risk and repayment behaviors, and what attributes define these segments?
2. How do acquisition channels and onboarding strategies impact portfolio quality, customer lifetime value, and unit economics?
3. Which loan products, ticket sizes, and tenures deliver the strongest balance between growth and risk?
4. How can pricing, approval, or tenure strategies be tailored across segments to improve overall portfolio outcomes?
5. What performance metrics and views should senior leadership monitor to proactively manage risk and growth?



DATA DESIGN GUIDELINES

Participants are required to design and generate their own synthetic dataset that reflects a realistic digital lending ecosystem. The dataset should capture the full customer journey and allow meaningful analysis of risk, growth, and performance across the following areas:

DATA AREA	CORE INFORMATION TO INCLUDE	EXPECTED RELATIONSHIP
CUSTOMER PROFILE	Geography, income proxy, employment type, credit quality indicator	Weaker profiles tend to exhibit higher risk
LOAN & PRODUCT	Product type, loan size, tenure, pricing, origination risk grade	Higher risk may result in tighter terms or higher pricing
REPAYMENT BEHAVIOR	Payment timeliness, delinquency status, partial payments	Deteriorating payment behavior precedes default
BEHAVIORAL SIGNALS	Cash flow consistency, balance volatility, spending shocks	Higher volatility may result in higher delinquency likelihood
ACQUISITION	Channel, cost of acquisition, approval turnaround	Faster/cheaper growth may carry higher risk
TIME DIMENSION	Origination and repayment periods	Newer cohorts may behave differently
OUTCOMES	Loan status, default indicator, value proxy	Certain segments outperform on risk-adjusted basis

DELIVERABLES

The final submission consists of two components: a **Credit Policy Recommendation Report** and a **Presentation deck**.

1. The report should be 4 to 6 pages in length, addressed to a fictional Chief Risk Officer. It must include an executive summary, risk segmentation findings, early warning signal logic, and a specific policy recommendation accompanied by a projected impact assessment. This format mirrors what a strategy analyst at a fintech or a consulting associate on a financial services engagement would be expected to produce.
2. The presentation deck should complement the report and communicate the team's solution to a stakeholder audience. Its structure and design are intentionally left to the participants' discretion; we are more interested in the quality of thinking than in adherence to a prescribed format.

Across both components, the standard being applied is this: the work should lead with strategic recommendations that are substantiated by data, not data outputs in search of a conclusion.

POINT OF CONTACT

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